ALMA Proposal Preparation Workshop

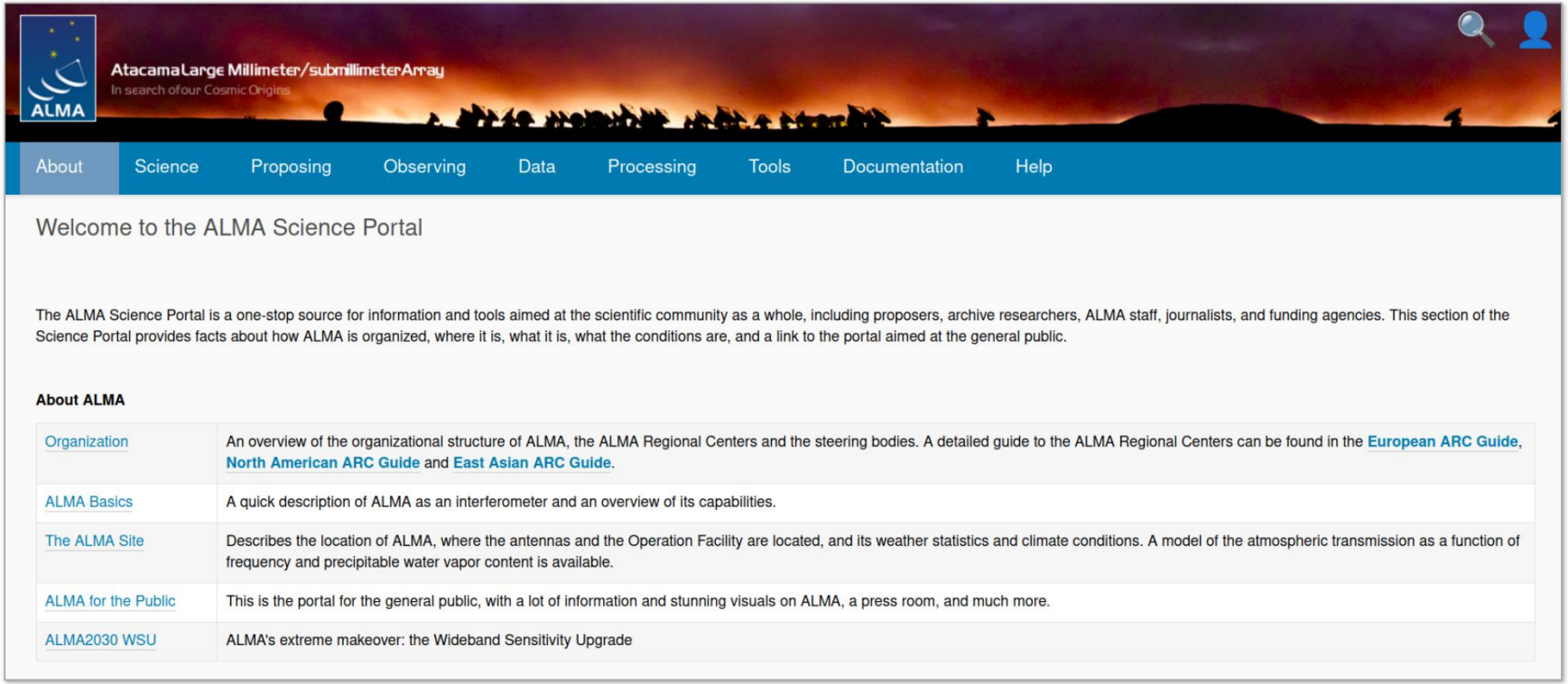
March 20, 2026 @IFA-UV, Valparaíso

# 2.1. ALMA Cycle 13 Capabilities

**Carla Cornil-Baïotto**

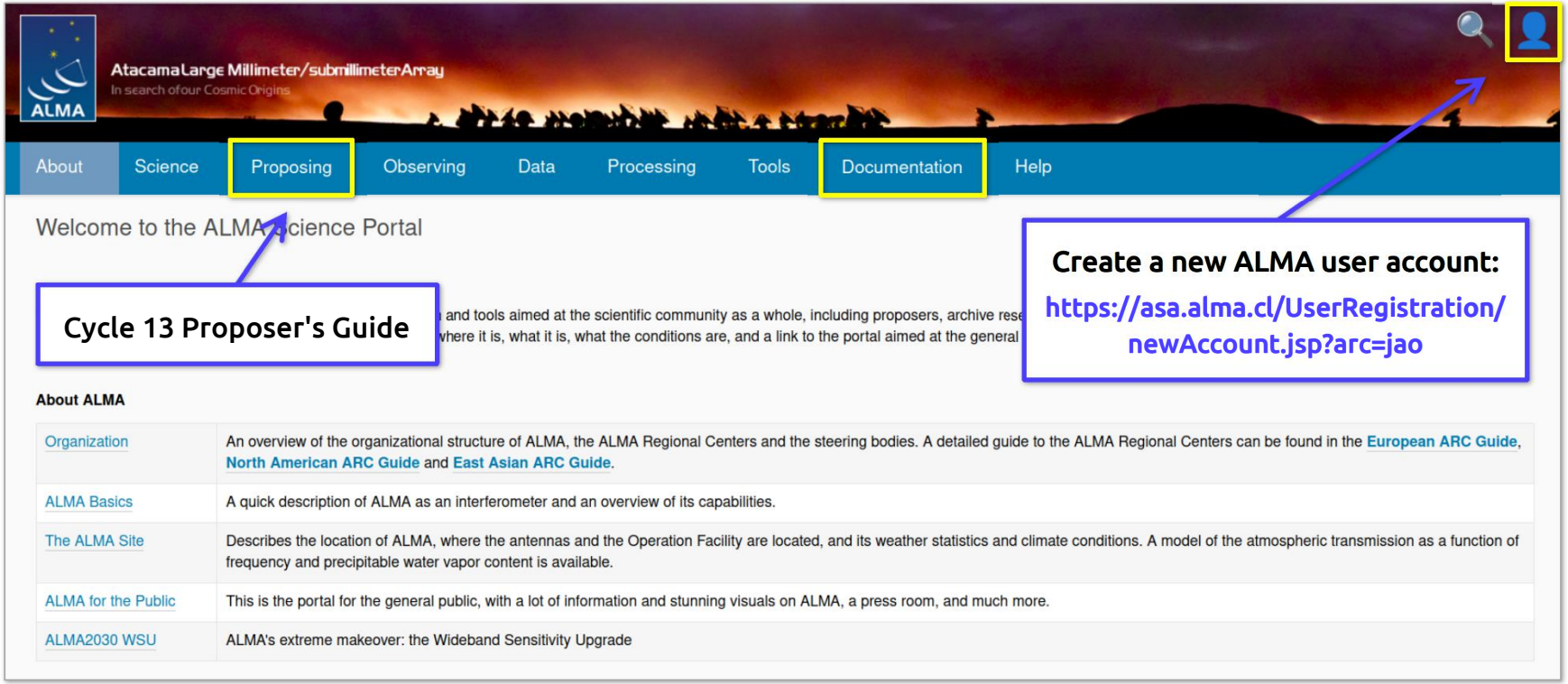
carla.cornil@postgrado.uv.cl

# ALMA Science Portal




**Link:** https://almascience.nrao.edu/about-alma

# ALMA Science Portal



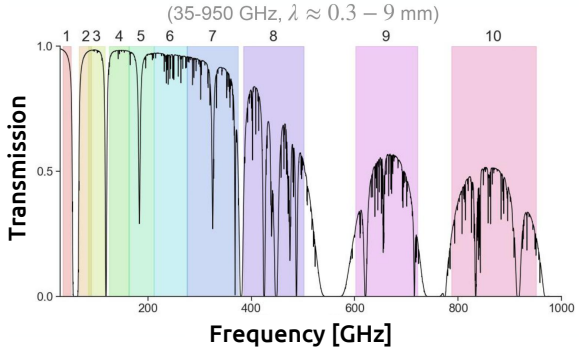

**Cycle 13 Proposer's Guide**

**Create a new ALMA user account:**
https://asa.alma.cl/UserRegistration/newAccount.jsp?arc=jao

**Link:** https://almascience.nrao.edu/about-alma

# ALMA Current Status

10 receiver bands




- ALMA Construction Project ended in September 2014
- Successful Long Baseline Commissioning Campaigns from 2014 to 2015 → Up to ~16 km baseline (C-10)
- 66 antennas: 54 × 12-m (incl. 4 for TP) + 12 × 7-m antennas
  - 12-m Array: minimum of 43 antennas and potentially up to 50 (usually maximum of 46-47) antennas are used for PI science observations
  - 7-m Array: minimum of 10 antennas and up to 12 antennas
  - TP Array: minimum of 3 antennas for single-dish maps with the ACA
  - Antennas not used for PI science are usually undergoing maintenance
- Cycle 13: All ten bands are now offered (incl. new Band 2)

# Cycle 13 Array Configurations

| Start date | Configuration | Max Baseline |
|---|---|---|
| 1 Oct 2026 | C-3 | 0.50 km |
| 20 Oct 2026 | C-2 | 0.31 km |
| 10 Nov 2026 | C-1 | 0.16 km |
| 30 Nov 2026 | C-2 | 0.31 km |
| 20 Dec 2026 | C-3 | 0.50 km |
| 10 Jan 2027 | C-4 | 0.78 km |
| 1 Feb 2027 | Maintenance | Maintenance |
| 1 Mar 2027 | C-4 | 0.78 km |
| 20 Mar 2027 | C-5 | 1.4 km |
| 20 Apr 2027 | C-6 | 2.5 km |
| 20 May 2027 | C-7 | 3.6 km |
| 20 Jun 2027 | C-8 | 8.5 km |
| 11 Jul 2027 | C-9 | 13.9 km |
| 30 Jul 2027 | C-10 | 16.2 km |
| 20 Aug 2027 | C-9 | 13.9 km |
| 10 Sep 2027 | C-8 | 8.5 km |

ALMA antenna transporter

Credit: NRAO

One day to move an antenna

- Configurations: **C-1 to C-10**
- Maximum baselines: from **0.16 km** to **16 km**

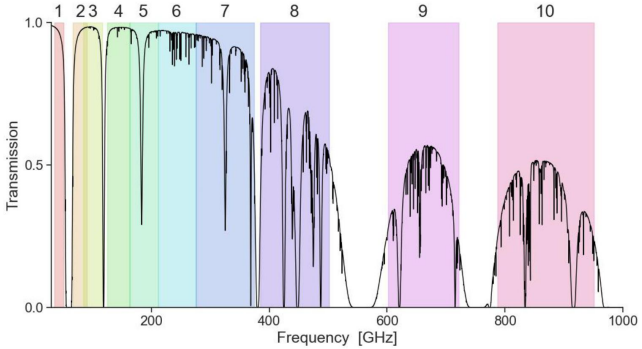
https://almascience.nrao.edu/observing/observing-configuration-schedule/long-term-configuration-schedule

- Files with ***representative*** antenna configurations for the 12-m and 7-m Arrays suitable for CASA simulations are available in the Science Portal

https://almascience.nrao.edu/tools/casa-simulator

## 10 receiver bands

(35-950 GHz, $\lambda \approx 0.3 - 9$ mm)

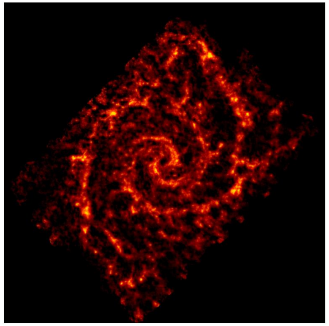



## Baselines: 0.16 to 16 km

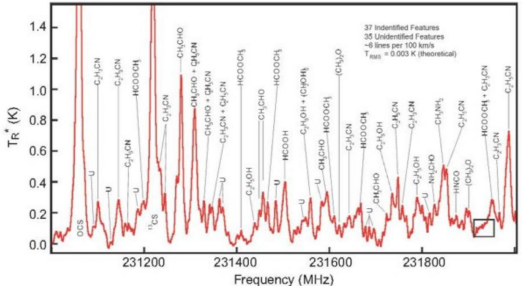
(resolution as fine as 0.005″ at 950 GHz)

## Imaging

High-fidelity imaging

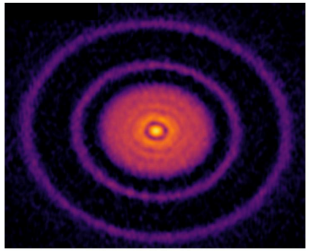


## Spectroscopy



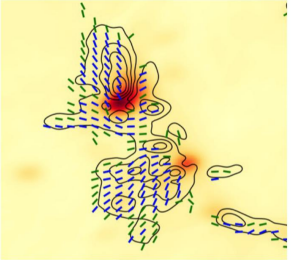

Sensitive spectroscopy to study molecular gas and chemistry

## Continuum



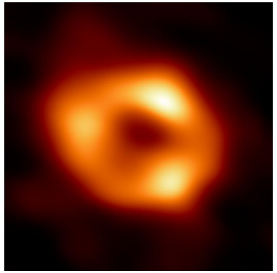
Trace dust and SF

## Polarization



Probe magnetic fields

## VLBI



EH scale resolution

Slide courtesy of John Carpenter

# ALMA Cycle 13 Capabilities

| Observing mode | 12-m Array | 7-m Array | TP Array |
|---|---|---|---|
| Spectral line imaging | Bands 1–10 | Bands 1 and 3–10 | Bands 1 and 3–10 |
| Continuum imaging | Bands 1–10 | Bands 1 and 3–10 | Bands 1 and 3–10 |
| Spectral scan mode | Bands 1–10 | Bands 1 and 3–10 | Bands 3–8 |
| New Band 2 (67–116 GHz) | Yes | No | No |
| Mosaic observations | Bands 1–10 | Bands 1 and 3–10 | N/A |
| Polarization | Full polarization (linear + circular), Bands 1–7 | Linear only, Bands 1 and 3–7 | Not supported |
| Band-to-band phase calibration (B2B) | Available (Bands 7–10, conditional on calibrator) | Available (Bands 7–10, conditional on calibrator) | Not available |

# ALMA Cycle 13 Capabilities

ALMA (ESO/NAOJ/NRAO)

## *WHAT'S NEW IN CYCLE 13*

- Band 2 (67–116 GHz) on the 12-m Array, including single-pointing polarization.
- Full polarization in Band 1 on the 7-m Array (single pointing), with observing performance comparable to Bands 3–7.
- Improved active phasing for VLBI: flux density threshold reduced by √8 vs. Cycle 12.
- B2B calibration now allowed for Large Programs (subject to LST constraints).

# ALMA Cycle 13 Capabilities

## ALMA Proposal Types

- **Regular Program** (PID 2026.1.xxx.S)
  Time limit: ≤ 50 h (12-m Array)
  ≤ 150 h (7-m stand-alone)
- **Very Long Baseline Interferometry (VLBI)**
  ALMA used as part of a global VLBI network
- **Phased Array**
  Antennas phased together as a single dish
- **Target of Opportunity (ToO)**
  Triggered by unpredictable/transient events
- **Large Program** (PID 2026.1.xxx.L)
  Major, legacy-type projects
  > 50 h (12-m) or > 150 h (7-m)

## Joint Proposals

| Facility | Available Time |
|---|---|
| JWST | 115 h /cycle |
| VLA | 50 h /year |
| VLT | 50 h /year |

- ALMA has agreements with JWST, VLA and VLT
- Science review by the ***main*** observatory only, plus a technical/scheduling review by all observatories

# ALMA Cycle 13 Capabilities

## Summary

- CfP: **March 19** to **April 23** (5 weeks)
- All receiver bands will be offered, including new **Band 2** (for the 12-m array only)
- The OT choice between Band 2 and 3 will be described in the Cycle-13 Proposer's Guide
- Largest configurations **C-9** & **C-10** offered
- Cycle 13 is not impacted by the WSU
- New web-based Observing Tool version (for the Phase-1 submissions only)

## Sources of Support

- ALMA Helpdesk

  Questions are usually answered within 48 hours
  Submit a ticket: https://help.almascience.org

- Face-to-face Visitor Support

  The NRAO/NAASC offers face-to-face meetings (online or in-person) for proposal preparation and data reduction support (PI or archival data)
  https://science.nrao.edu/facilities/alma/visitors-shortterm